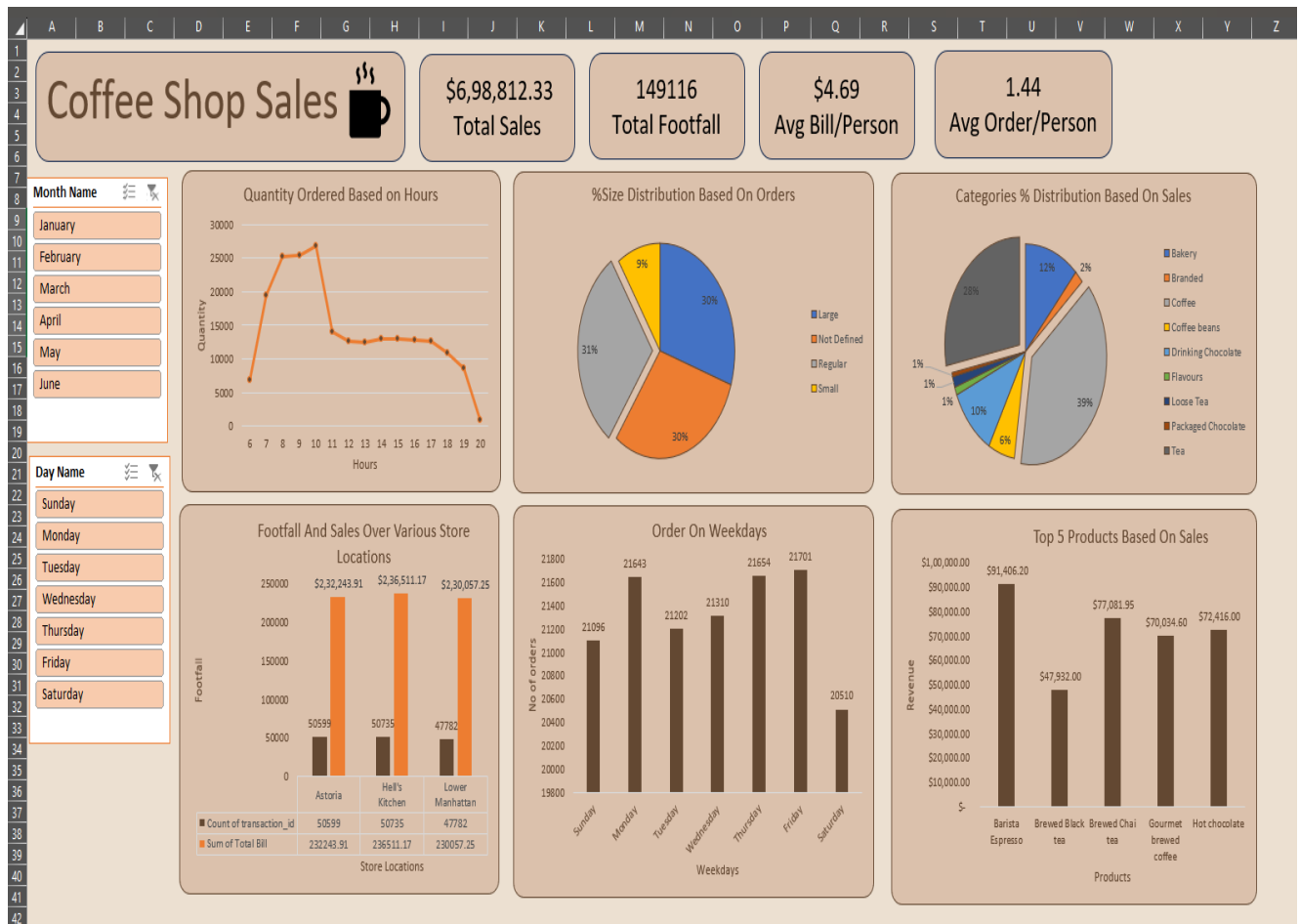


Coffee Shop Sales Analysis Project – Final Report



1. Business Questions Solved:

- How do sales vary by day of the week and hour of the day?
- Are there any peak times for sales activity?
- What is the total sales revenue for each month?
- How do sales differ across store locations?
- What is the average bill and number of orders per customer?
- Which products are best-selling in terms of quantity and revenue?
- How do sales vary by product category and product size?

2. Steps Involved:

Data Preparation:

- Imported raw transaction data (149,116 records, 11 fields) into Excel.
- Used **Power Query Editor** to clean data:
 - Removed duplicate transaction IDs.
 - Formatted date and time fields.
 - Added new calculated columns:
 - *Total Bill = Quantity × Unit Price*
 - *Month Name, Day Name, Hour of Day*
 - *Product Size* (categorized as Small, Regular, Large, Not Defined)

Data Analysis:

- Built **pivot tables** to explore patterns:
 - Sales by day, hour, and month.
 - Quantity ordered by hour.
 - Top 5 products by revenue.
 - Sales by store location and product category.
- Calculated key **KPIs**:
 - Total Sales, Footfall, Average Bill per Person, Orders per Person

Visualization:

- Created meaningful charts:
 1. Line Chart – Hour vs Quantity Ordered
 2. Column Chart – Store vs Footfall & Revenue
 3. Column Chart – Orders by Weekday
 4. Column Chart – Top 5 Products by Sales
 5. Pie Chart – Product Size Distribution
 6. Pie Chart – Sales by Product Category

Dashboard Development:

- Designed a fully interactive Excel dashboard using slicers for **Month Name** and **Day Name** to filter insights dynamically.
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3. Summary:

This project involved analyzing a large dataset of coffee shop transactions to uncover actionable insights that could support data-driven decisions. Using Excel and Power Query, the data was cleaned, transformed, and enriched with new fields. Pivot tables and visualizations were then used to identify sales patterns across time, location, and product types. An interactive dashboard was developed to allow dynamic exploration of the data.

Outcome: The analysis helped the business understand customer behavior, peak operational hours, top-performing products, and store performance—supporting better inventory planning, marketing strategy, and resource allocation.